TDI / insertion pipeline: toy-model report

What the toy models are

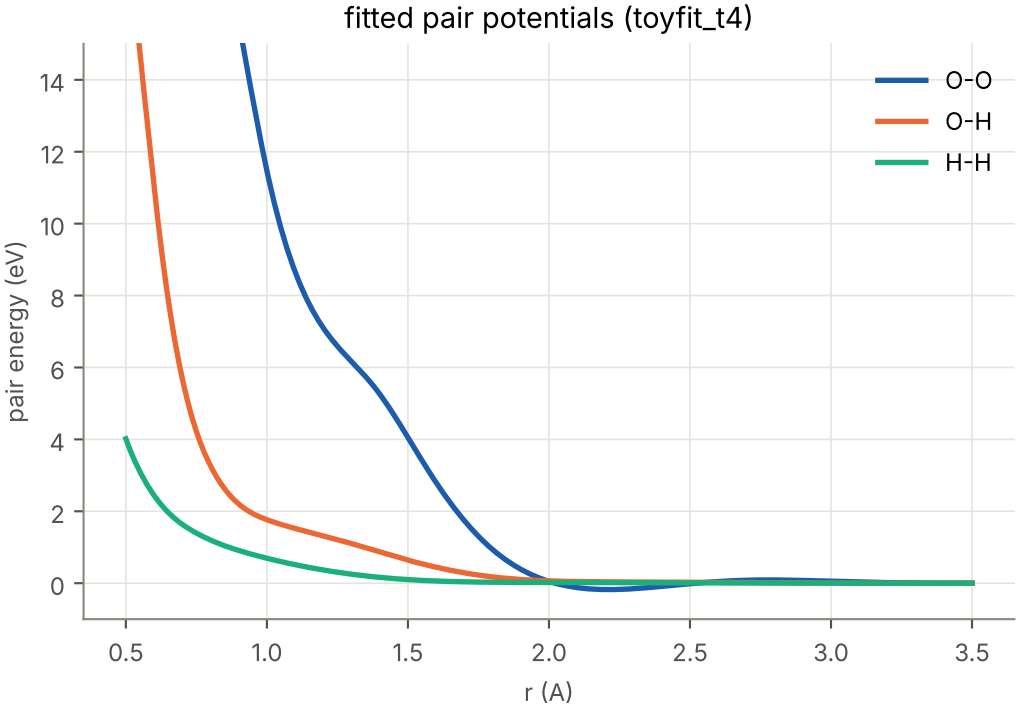
Pair potentials that run the unchanged production pipeline (run_real.py, young_tdi_par.sh) on CPU: 'analytic toy' (Born-Mayer O-O, Morse O-H, soft H-H) and 'fitted toy' (spline pairs force-matched to the DFT frames, toyfit_t4). Both give a superionic fcc H2O at 400-500 GPa (O vibrates, H diffuses). They are cheap enough for independent references (Widom / deletion / BAR) and for long runs, so they test the METHOD, not GRACE.

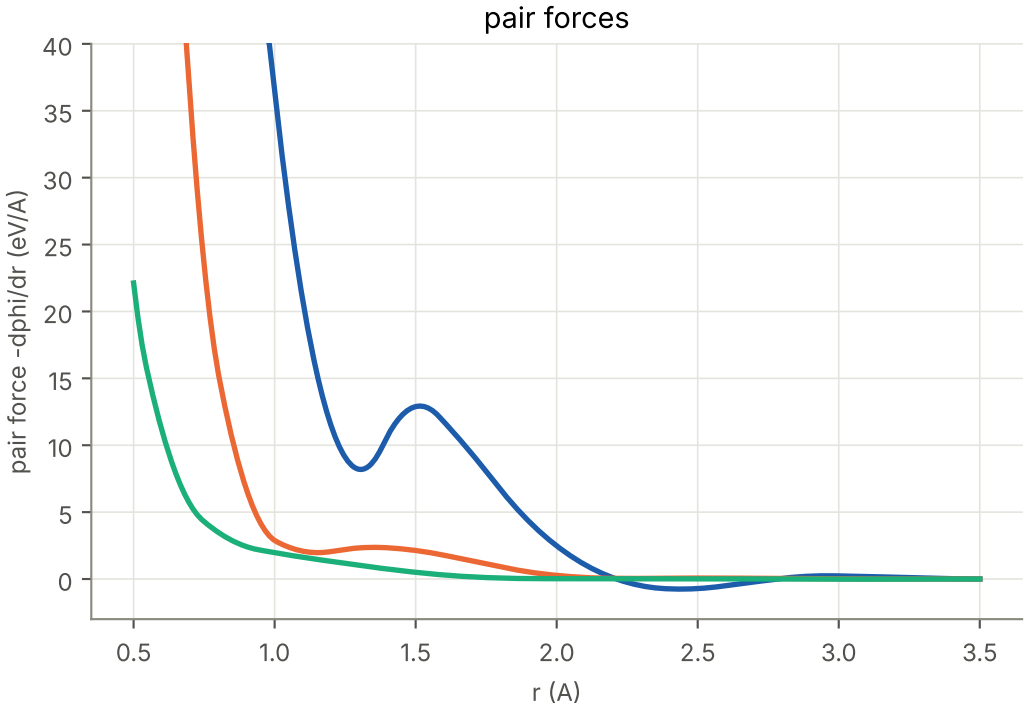
Findings

1. Bookkeeping is exact (T0): crystal route = insertion route = analytic formulas to 1e-13 eV. No kT-sized offset is possible.
2. The lambda integral agrees with independent routes: method check (insertion into pure H = G_H) passes at 3000 / 5000 K, insertion n = 8 = crystal route, insertion = Widom/BAR within ~1-2 sigma, MBAR = TI. 10 Gauss lambda values are too coarse for crystals (-14 meV/atom).
3. Quick checks (450 GPa, 5000 K): path independence (x^2 vs x^4), finite size (108 vs 32 O), linearity in composition: all within 1 sigma.
4. Scan of dg = mu_H(ice) - G_H over 11 solid (P, T) states: dg > 0 when cold, < 0 when hot; strongly P dependent (300 GPa: +530..+550, 400 GPa: +215 at 2000 K → -207 at 6000 K, 500 GPa: -86 at 4000 K → -416 at 7000 K). Insertion becomes favourable with T and with P. The pipeline and Widom/BAR agree on this T trend in the toy (T5 slope -0.069 vs -0.058 meV/K), so the method does not flip it.
5. GRACE: dg -300..-600 meV at 5000-7000 K (favourable), dg/dP -1.5 meV/GPa (toy: about -3); its T slope is not resolved (+80 +- 26 per 1000 K with all points, +16 +- 36 at 500 GPa only). The toy says the sign of the T trend should be negative; GRACE needs lower-T points to test it.

Caveats

The fitted toy errs by 40 meV/atom and 1.5 eV/A: its dg values are not predictions for water, only the trends and the method checks carry over.
Toy ice melts at a lower T than GRACE ice (400 GPa: between 6000 and 7000 K; 300 GPa: below 4000 K). 300 GPa and T < 4000 K are outside its fit data.
Missing: 500 GPa 2000/3000 K and 600 GPa (ghost placement in dense pure H, fixed; rerun pending).



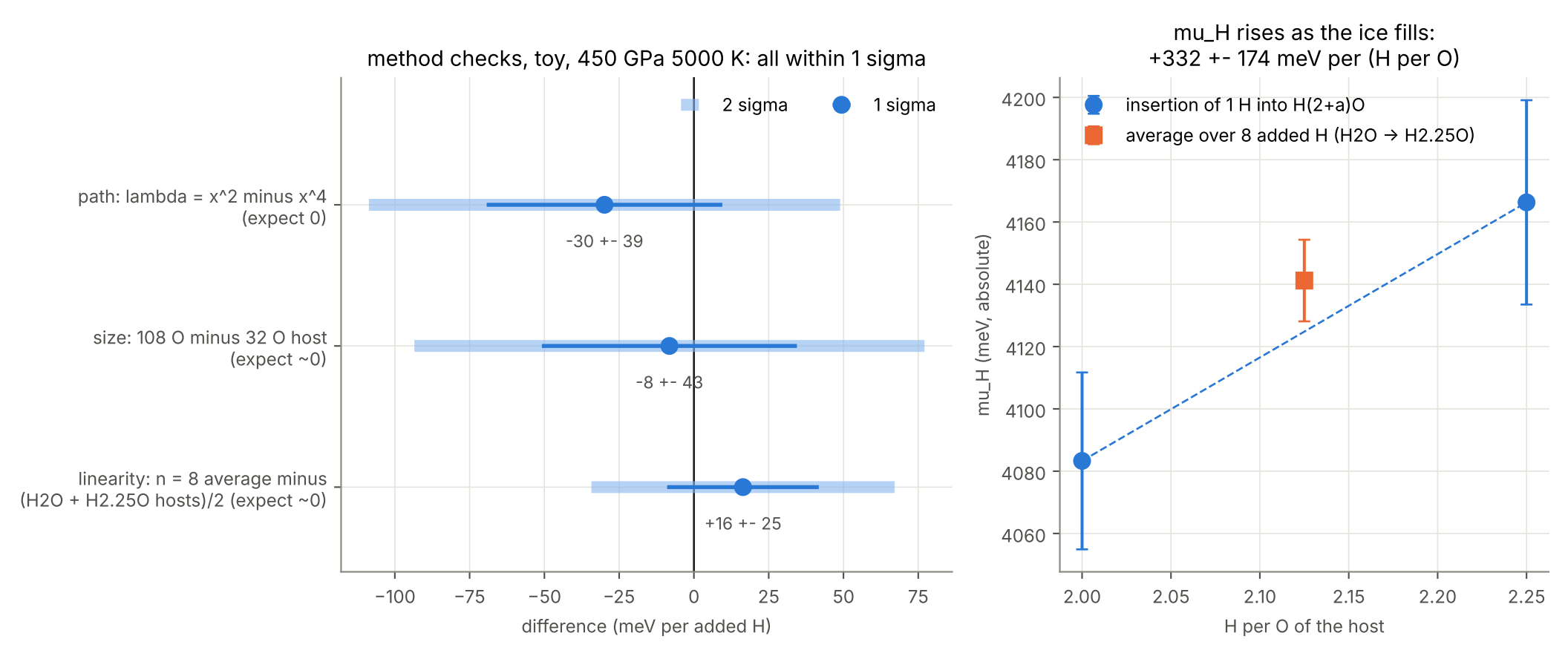


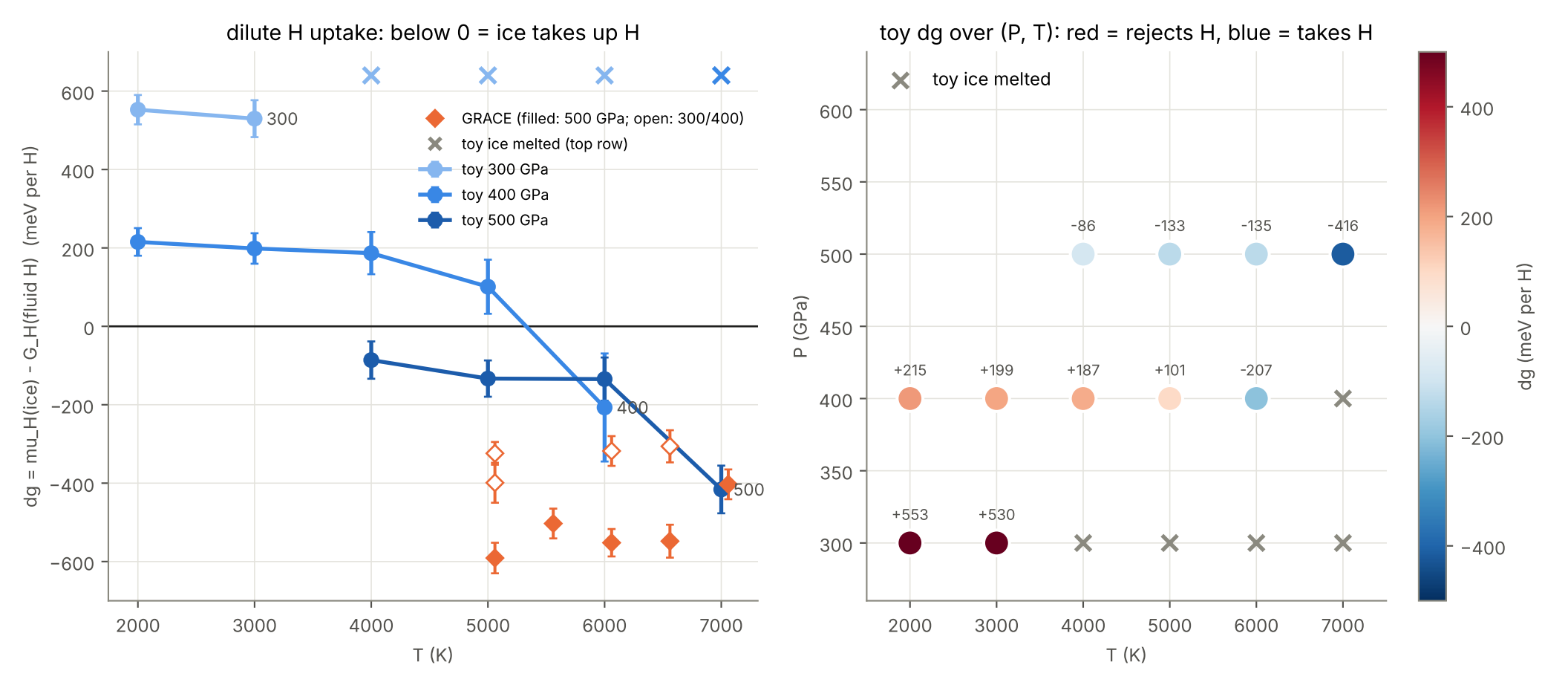
Force-matched to 341 DFT frames (250-700 GPa, T >= 4000 K; frames stay in scratch, not in this public repo). Errors train / validation: energy MAE 40 / 44 meV/atom, force RMSE 1.53 / 1.64 eV/A (DFT force rms 3.4), stress MAE 7.4 / 8.4 GPa.

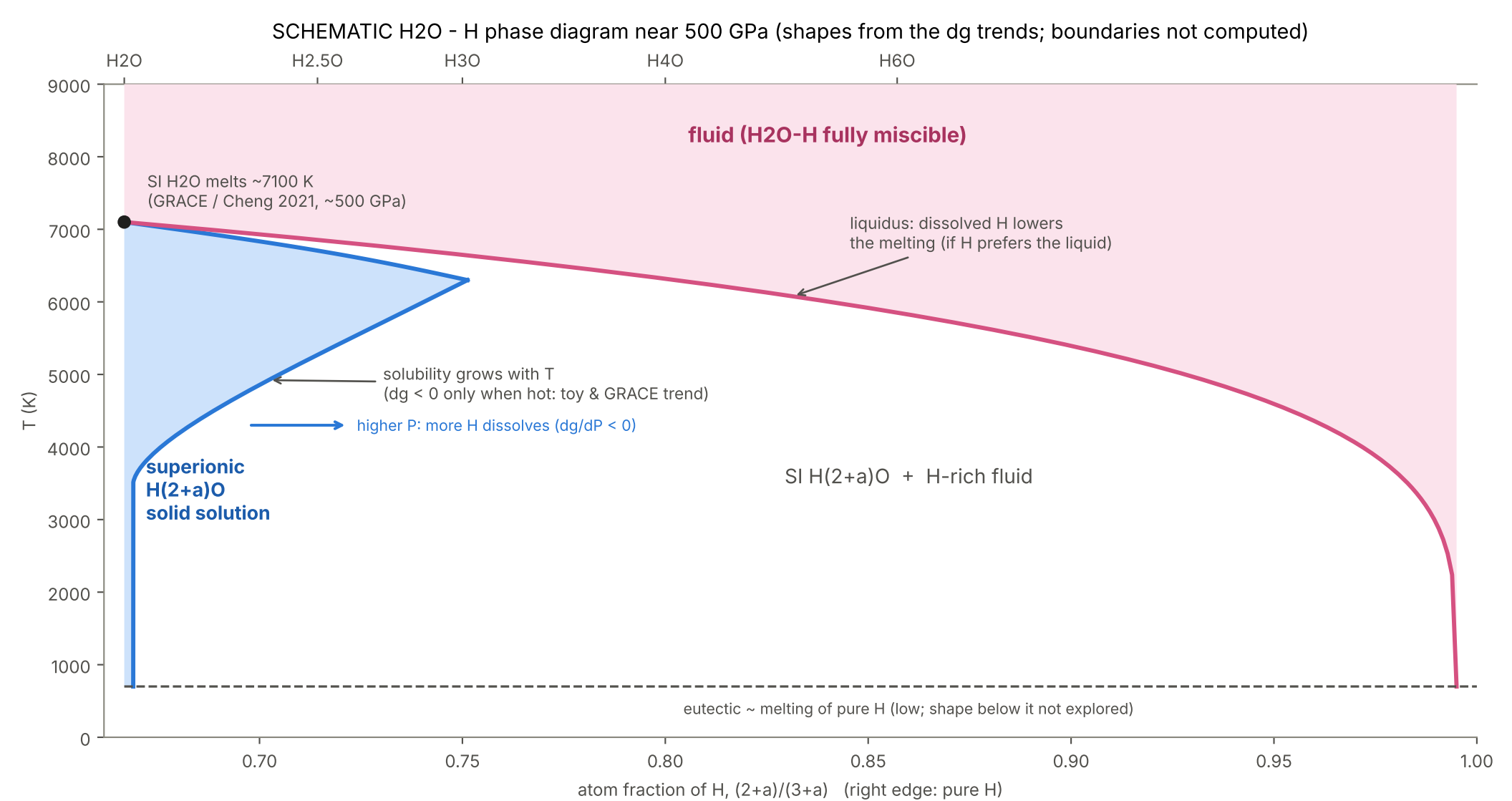
Pipeline validation on the toy models (meV per added H unless noted)

test	model	result	+-	verdict
T0 bookkeeping: crystal route = insertion route = exact formulas	analytic toy	exact to 1e-13 eV		PASS
T3 method check: insertion into pure H - G_H - finite-size term, 3000 K	analytic toy	-6.5	15	PASS
T3 method check, 5000 K	analytic toy	-3.4	28	PASS
T4 insertion n=8 - crystal route [G(H2.25O) - G(H2O) - 8 G_H]/8, 21 lambda	analytic toy	+10	57	PASS
T1 insertion integral - Widom/BAR (H2O host), 3000 / 5000 K	analytic toy	+45 / +36	40 / 29	PASS
T2 insertion integral - Widom/BAR (pure-H host), 3000 / 5000 K	analytic toy	-16 / -40	23 / 28	PASS
T1 insertion integral - Widom/BAR (H2O host), 5000 / 6000 K	fitted toy	+4.5 / +5.1	15.6 / 17.6	PASS
T2 insertion integral - Widom/BAR (pure-H host), 5000 K	fitted toy	+12.1	15.3	PASS
T3 method check, 5000 K (expected -7.9)	fitted toy	+1.7 vs expected	16.8	PASS
T5 dg pipeline - dg Widom/BAR, 5000 / 6000 K	fitted toy	+41.9 / +31.1	24.6 / 31.1	PASS (1.7 / 1.0 sigma)
T5 slope d(dg)/dT 5000-6000 K: pipeline -0.069, Widom -0.058 meV/K	fitted toy	-0.011	0.040	PASS
MBAR over the lambda series - TI (21-lambda insertion runs)	analytic toy	-39 .. +7	12-26	within errors
Gauss-Kronrod - Gauss quadrature, H2.25O crystal (per atom)	analytic toy	-14.1		use 21 lambda

T0 is exact (no statistics): the reference free energies (Einstein crystal, ideal-gas H, ln N!, thermal wavelength, PV, ghost placement) of the crystal route and the insertion route are identical, so no kT-sized bookkeeping offset is possible. T3/T4 are two-route checks of the lambda integral. Widom/BAR references (T1, T2, T5) are themselves only good to ~50 meV in the dense host (4 → 12 ps moved them by up to 80 meV).


method checks, toy, 450 GPa 5000 K: all within 1 sigma
2 sigma
1 sigma
path: lambda = x^2 minus x^4
(expect 0)
-30 +- 39
size: 108 O minus 32 O host
(expect ~0)
-8 +- 43
linearity: n = 8 average minus
(H2O + H2.25O hosts)/2 (expect ~0)
+16 +- 25
−100
−75
−50
−25
0
25
50
75
difference (meV per added H)
mu_H rises as the ice fills:
+332 +- 174 meV per (H per O)
insertion of 1 H into H(2+a)O
average over 8 added H (H2O → H2.25O)
mu_H (meV, absolute)
4200
4180
4160
4140
4120
4100
4080
4060
2.00
2.05
2.10
2.15
2.20
2.25
H per O of the host


dilute H uptake: below 0 = ice takes up H
dg = mu_H(ice) - G_H(fluid H) (meV per H)
T (K)
GRACE (filled: 500 GPa; open: 300/400)
toy ice melted (top row)
toy 300 GPa
toy 400 GPa
toy 500 GPa
300
400
500
toy dg over (P, T): red = rejects H, blue = takes H
toy ice melted
P (GPa)
T (K)
dg (meV per H)
-86
-133
-135
-416
+215
+199
+187
+101
-207
+553
+530



Built from: toy scan (dg > 0 cold, < 0 hot; more negative at higher P), toy checks (mu_H rises ~330 meV per H/O as the ice fills: finite solubility), GRACE dg < 0 at 5000-7000 K, GRACE liquid hull (miscible). Not computed: solidus / liquidus positions, the partition coefficient (needed for the sign of the melting-point change), the eutectic.